

Principal square roots

The principal square root is the nonnegative number whose square equals the radicand.

$$\sqrt{a}, \quad a \geq 0$$

Why is the principal square root of 36 equal to 6 rather than -6?

Evaluating a square root

$$\sqrt{\frac{49}{64}}$$

What is the exact value?

Square roots and squares

$$\sqrt{x^2}$$

What expression is equivalent for every real x ?

A negative input

$$\sqrt{(-5)^2}$$

What value does this expression have?

Even roots

An even root of a negative number is not a real number.

$$\sqrt[4]{-16}$$

Does this expression have a real value?

Odd roots

$$\sqrt[3]{-125}$$

What is the real value?

Product rule for square roots

Assume a and b are nonnegative.

$$\sqrt{ab} = \sqrt{a}\sqrt{b}$$

How can this rule help simplify the square root of 72?

Simplifying a radical

$$\sqrt{108}$$

What simplified radical form gives the exact value?

Variable radicals

Let x be any real number.

$$\sqrt{50x^2}$$

What simplified expression accounts for both positive and negative x ?

Quotient rule for square roots

Assume a is nonnegative and b is positive.

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

Why does the denominator require a stricter condition?

A radical quotient

$$\sqrt{\frac{18}{25}}$$

What is the simplified exact value?

Combining radicals

$$4\sqrt{5} - \sqrt{5} + 2\sqrt{5}$$

What equivalent expression combines these terms?

Simplifying before combining

$$\sqrt{48} + \sqrt{75}$$

What is the simplified exact sum?

A radical of a sum

$$\sqrt{9 + 16}$$

Is its value equal to the sum of the square roots of 9 and 16?

Multiplying radicals

$$\sqrt{6}\sqrt{15}$$

What is the simplified exact product?

Rationalizing a denominator

$$\frac{7}{\sqrt{3}}$$

What equivalent fraction has a rational denominator?

Conjugates

Conjugates have the same terms with the sign between them reversed.

$$2 + \sqrt{3} \quad \text{and} \quad 2 - \sqrt{3}$$

What happens when we multiply these conjugates?

Rationalizing a binomial denominator

$$\frac{4}{1 + \sqrt{5}}$$

What equivalent expression has a rational denominator?

Rational exponents

For nonnegative a and positive integer n :

$$a^{1/n} = \sqrt[n]{a}$$

What does an exponent of one-fourth mean?

A fractional power

$$27^{2/3}$$

What is the exact value?

A negative fractional power

$$16^{-3/4}$$

What is the exact value?

A variable fractional power

Assume x is nonnegative.

$$x^{3/2}$$

How can we write this expression using a radical?

Simplifying a higher root

Let x be any real number.

$$\sqrt[3]{54x^4}$$

What simplified radical expression is equivalent?